```
import numpy as np
import os
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.ensemble import IsolationForest
from sklearn.metrics import confusion_matrix, classification_report
```

Build different Normalities

```
def load_timeseries_from_folders(normality, folder_list):
    all_dfs = []
    folders_to_read = folder_list[:normality]

    for folder in folders_to_read:
        # Full folder path
        folder_path = os.path.join("data", "TSB-UAD-Public", folder)

        # List all files in the folder
        files = [f for f in os.listdir(folder_path) if
os.path.isfile(os.path.join(folder_path, f))]

        if not files:
            print(f"Folder {folder} is empty or contains no files.")
            continue

        # Use the first file found
        filename = files[0]
        file_path = os.path.join(folder_path, filename)

        try:
            df = pd.read_csv(file_path, header=None).dropna()
            df.columns = ['measurement', 'anomaly']
            print(f"Read file {filename} from folder {folder}
({df.shape[0]} rows).")
            all_dfs.append(df)
        except Exception as e:
            print(f"Error reading file {file_path}: {e}")

    if all_dfs:
        full_df = pd.concat(all_dfs, ignore_index=True)
        print("\nData merge completed. Combine df
has",full_df.shape[0]," number of rows")
        return full_df
    else:
        print("No valid data found.")
        return pd.DataFrame()

# Example usage
```

```
folders = ["YAHOO", "SensorScope", "GHL", "Genesis"]
normality = 1 # increment up to 4 , for different normality datasets

result = load_timeseries_from_folders(normality, folders)
result.to_csv('normality1.csv',index=False)
# result.to_csv('normality2.csv',index=False)
# result.to_csv('normality3.csv',index=False)
# result.to_csv('normality4.csv',index=False)
result
```

```
Read file YahooA3Benchmark-TS100_data.out from folder YAHOO (1680
rows).
Read file stb-10.test.out from folder SensorScope (29518 rows).
Read file 01_Lev_fault_Temp_corr_seed_11_vars_23.test.csv@10.out from
folder GHL (200001 rows).
Read file genesis-anomalies.test.csv@1.out from folder Genesis (16220
rows).

Data merge completed. Combine df has 247419  number of rows
```

	measurement	anomaly
0	6.470061	0
1	80.972293	0
2	159.014680	0
3	214.721803	0
4	201.695834	0
...	...	...
247414	8.000000	0
247415	-21.000000	0
247416	-17.000000	0
247417	-17.000000	0
247418	-27.000000	0

```
[247419 rows x 2 columns]
```